Examination Bioinformatics

Linköpings Universitet, IDA, Statistik

Course:	732A51 Bioinformatics
Date:	2026/10/01, 8–12
Assisting teacher:	Bayu Beta Brahmantio
Examinator:	Krzysztof Bartoszek
Provided aids:	The help material is included in the zip file **exam_help_material_732A51.zip**.
Grades:	A= $[18-20]$ points B= $[16-18)$ points C= $[14-16)$ points D= $[12-14)$ points E= $[10-12)$ points F= $[0-10)$ points
Instructions:	Provide a detailed report that includes plots, conclusions and interpretations. Give motivated answers to the questions. If an answer is not motivated, the points are reduced. Provide all necessary codes in an appendix. If you are asked to do plots, then make sure that they are informative, have correctly labelled axes, informative axes limits and are correctly described. Points may be deducted for poorly done graphs. Name your digital part solution files as: **[your exam account id]_[own file descripti** If you have problems with creating a pdf you may submit your solutions in text files with unambiguous references to graphics and code that are saved in separate files There are **FOUR** assignments (with sub–questions) to solve. Include all code that was used to obtain your answers in your solution files. Make sure it is clear which code section corresponds to which question. Your code should be complete and readable, possible to run by copying directly into a script. Comment directly in the code whenever something needs to be explained or discussed. If you also need to provide some hand–written derivations please number each page according to the pattern: Question number . page in question number i.e. Q1.1, Q1.2, Q1.3,. . ., Q2.1, Q2.2, . . ., Q3.1, ...

Problem 1 (6p)

Use the standard genetic code table (in the slides of Lecture 1). The codon GAG codes for glutamic acid (E).

a) (2p) List all nine single-nucleotide substitutions of GAG, and what this produces, and classify each as synonymous, or nonsynonymous.

b) (2p) Repeat the classification for the codon TGG (tryptophan, W). Which codon, GAG or TGG, is more "fragile" with respect to nonsense mutations?

c) (2p) Explain why is it beneficial to have synonymous substitutions. Notice that synonymous substitutions are more likely at the third position, explain why this implies that when constructing a phylogeny the third codon position should be treated differently.

Problem 2 (6p)

A eukaryotic gene fragment (genomic DNA, coding strand, $5' \to 3'$) and the corresponding cDNA (reverse-transcribed from the mature mRNA, so introns are removed) are:

```
genomic: ATGGCTAAGGTATGCCTTAGGACCGTAAATTCTGA   (35 nt)
cDNA:    ATGGCTAAGGACCGTAAATTCTGA               (24 nt)
```

a) (2p) By comparing the two sequences, determine the exact position of the intron in the genomic sequence and write down the two exons and the intron.

b) (1p) Almost all introns start with GT and end with AG. Check whether your intron obeys this rule.

c) (1p) Translate the cDNA with the standard genetic code and give the protein sequence (table in Lecture 1's slides).

d) (2p) Translate the genomic sequence in the same reading frame as if the intron were *not* removed. What happens to the protein, and why is the effect of an un-spliced intron of length 11 different from one of length 12?

Problem 3 (4p)

A biologist has annotated two short DNA sequences with the true hidden states of the model in Question 8 (I = island, N = non-island):

```
sequence 1:       C G C A T A        sequence 2:       A T G C G T
(hidden) states:  I I I N N N        (hidden) states:  N N I I I N
```

a) (2p) Using this fully observed (supervised) data, write down the maximum-likelihood estimates (fractions) of the initial distribution π, the transition matrix P and the emission probabilities $b_I(\cdot)$, $b_N(\cdot)$.

b) (2p) Some of your emission estimates are exactly zero. Why is this a problem when the model is later used to decode new sequences? Re-estimate the emission probabilities adding a pseudocount of 1 to every (state, nucleotide) cell.

Problem 4 (4p)

Let x = AAAAAAAA (eight A's) and y = AATAATAATAA. Two alignments are proposed:

```
Alignment B:   A A - A A - A A - A A        Alignment A:   A A - - - A A A A A A
               A A T A A T A A T A A                       A A T A A T A A T A A
```

Score matches +1, mismatches −1. Consider (i) a linear gap penalty $\delta(l) = -l$ and (ii) an affine penalty $\delta(l) = -4 - (l-1)$ (gap open 4, gap extension 1).

a) (2p) Score both alignments under both penalty schemes (four numbers) and state which alignment is preferred under each scheme.

b) (2p) Give the biological motivation for affine gap penalties. What does a gap of length three imply if this was part of a gene?